



Open Project'26

**Agentic AI-Based Dynamic Tariff Optimization for EV
Charging Networks Using Large-Scale Charging Session
Data.**

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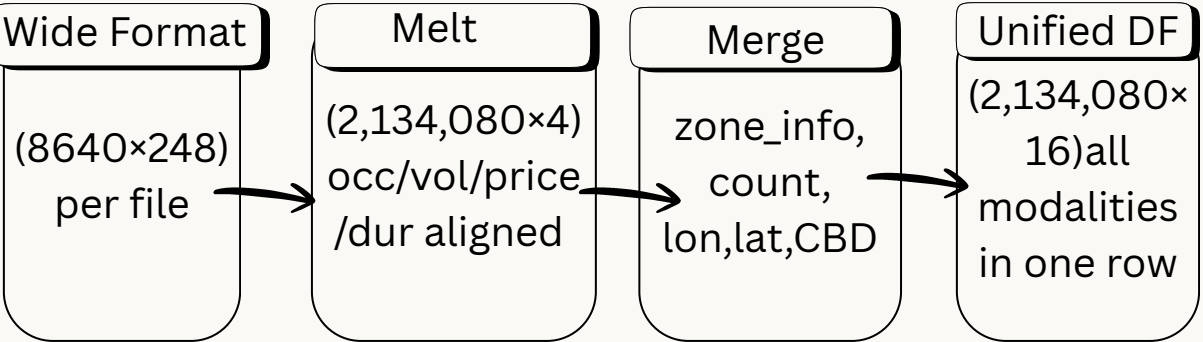
Urban EV Dataset (ST-EVCDP) Preprocessing & Feature Engineering:

247 Zones, 2.13M Records — From Raw Files to ML-Ready Signal

Dataset	Shape	Dtypes
occ_raw	(8640, 248)	248×i64
vol_raw	(8640, 248)	247×f64, 1×i64
price_raw	(8640, 248)	242×f64, 6×i64
dur_raw	(8640, 248)	247×f64, 1×i64
info_raw	(247, 10)	7×i64, 3×f64
time_raw	(8640, 6)	6×i64
adj_raw	(247, 248)	248×i64
dist_raw	(247, 248)	247×f64, 1×i64
stations	(1706, 6)	4×i64, 2×f64

Zero missing values across all 9 raw files — no imputation needed at ingestion stage

Wide-to-Long Transformation:



8,640 timestamps × 247 zones = 2,134,080 row

Multicollinearity Resolution: 75 columns → 25 final ML features

- neighbor_wt_util dropped — $r > 0.92$ with neighbor_avg_util; distance-weighted average adds no information over simple average
- util_roll_mean_6h dropped — $r > 0.85$ with both util_roll_mean_1h and util_ewm_1h; redundant trend signal
- minutes_since_midnight dropped — $r = 1.0$ with interval_of_day; identical linear transformation of the same variable

FEATURE ENGINEERING:

ECONOMIC AND DEMAND SIGNALS

utilization_rate	mean 0.2802	revenue	mean 29.99
CUR	mean 0.2257	queue_proxy	mean 0.0154
surge_flag	1.02% of records	discount_flag	60.85% of records

TEMPORAL FEATURES

hour_sin / hour_cos — mean 0.0	day_sin / day_cos
is_morning_rush, is_evening_rush	is_midnight_window 33.3% of rec
period split	off_peak 889K peak 622K shoulder 622K

LAG AND ROLLING FEATURES

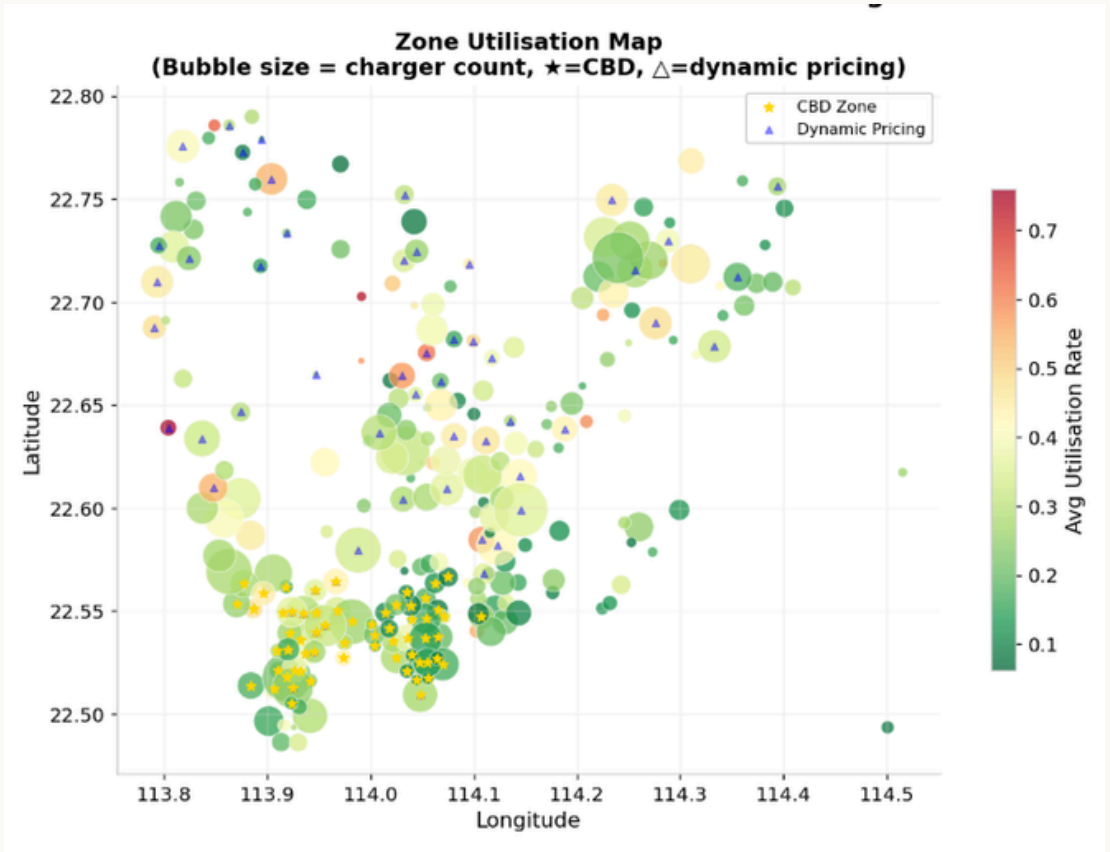
Lag lookbacks-util_lag_1 (5 min) util_lag_12 (1 hr) util_lag_288 (24 hr)
Rolling stats (1h window)- roll_mean roll_std roll_max ewm_1h
Delta features- util_delta_1 util_delta_12

SPATIAL FEATURES

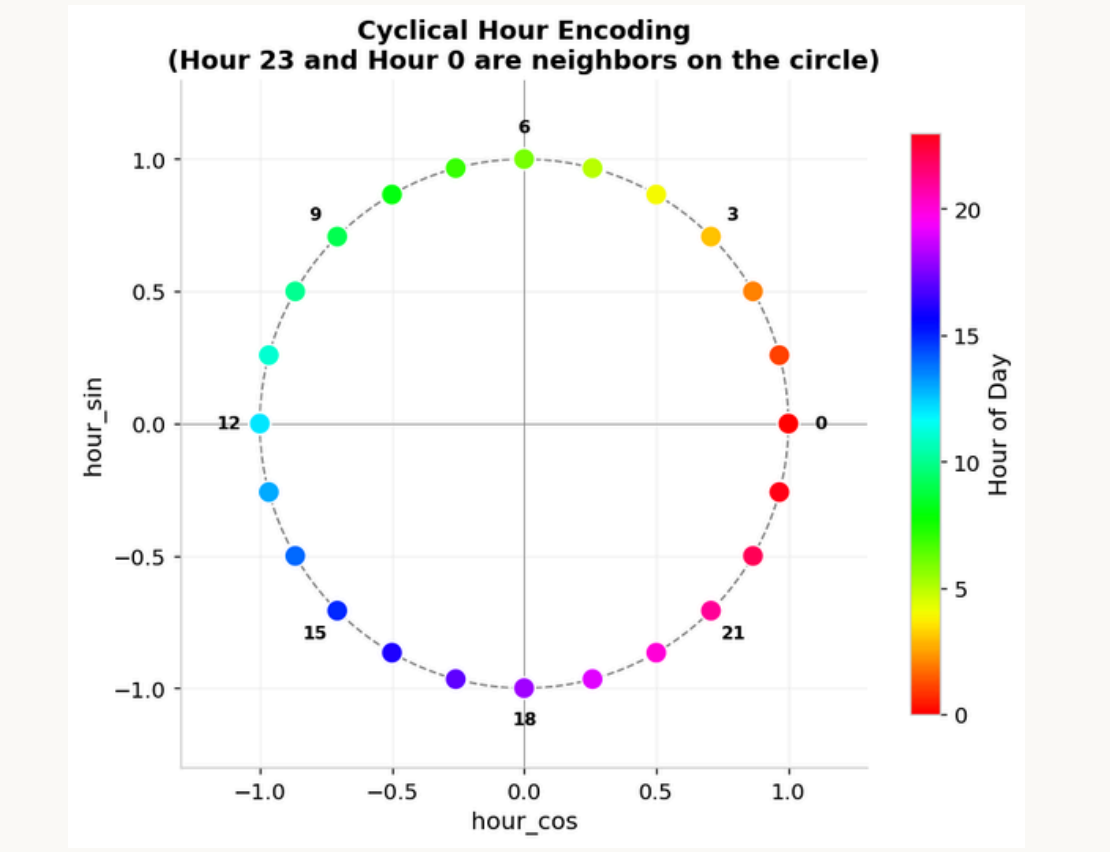
neighbor_avg_util, neighbor_max_util, spillover_pressure
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TARGET FEATURE

target_util = utilization_rate shifted by 1 step forward per zone



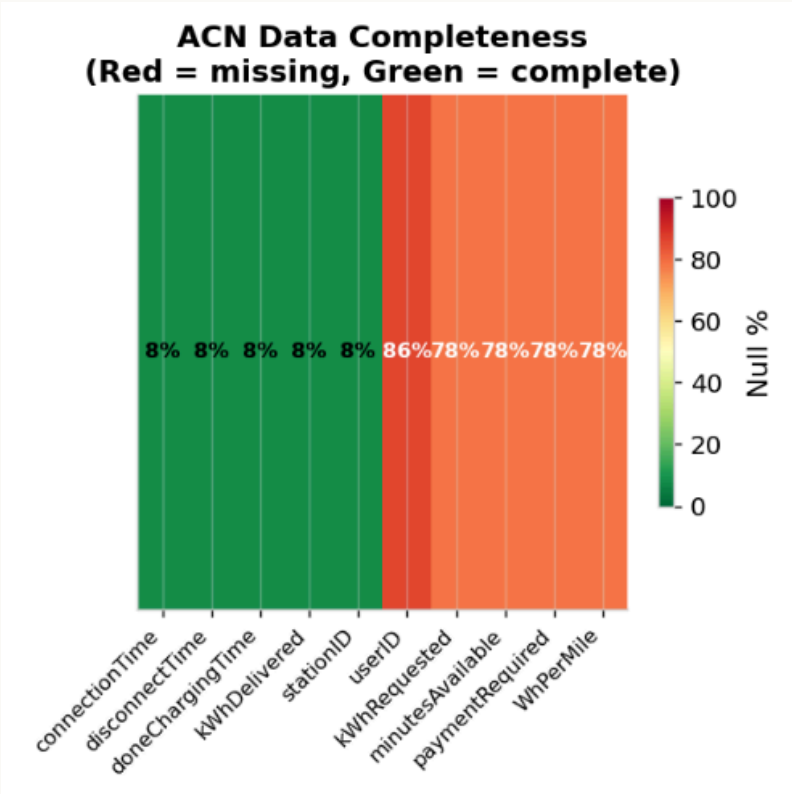
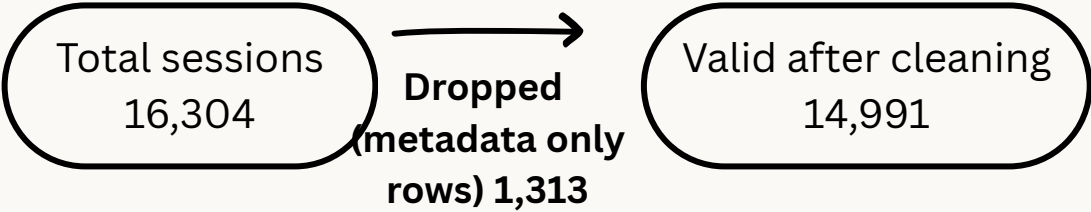
CBD zones (★) cluster in the south-west with the densest charger coverage; high-utilisation zones (yellow-red) concentrate in the north-east, signalling spatial mismatch between supply and demand.



Hour 23 and Hour 0 are adjacent on the circle — sin/cos encoding preserves this continuity that ordinal integers break.

ACN(Caltech) Data Preprocessing & Feature Engineering: (Apr 25 – Dec 16, 2018)

14,991 Real Charging Sessions – From Raw API Logs to Behavioural Features



100% of API scaffold columns (_meta, _items, end, start, userInputs) dropped – zero analytical value

Issue Found	Fix Applied
18 negative charge_hr values	abs() correction
327 charging rate outliers	Capped to [0.5, 100] kW
kWhRequested – 85.1% missing	Stratified median by hour × is_weekend
minutesAvailable – 78.3% missing	Median imputation
WhPerMile – 78.3% missing	Median imputation
11 API/metadata columns	Dropped entirely

FEATURE ENGINEERING:

SESSION-LEVEL SIGNALS:

- parked_hours –

total time at station
- charging_hours –

active charge time
- idle_hours –

parked but not charging
- idle_ratio –

mean 0.31

charging_rate_kw –

mean 3.63 kW
- session_hr –

mean 5.93 h

charge_hr –

mean 3.24 h

BEHAVIOURAL/TEMPORAL FLAGS:

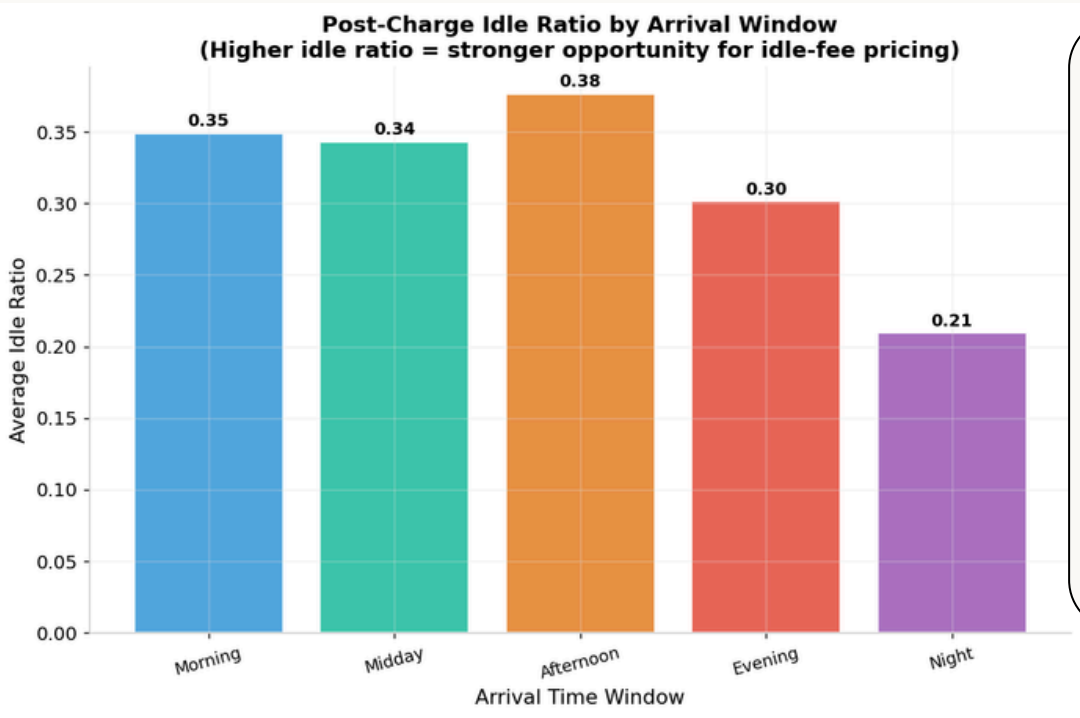
- arrival_bin – 5 categories

weekday_cat – ordinal encoding
- is_weekend – 2,796 weekend vs 12,195 weekday
- period split

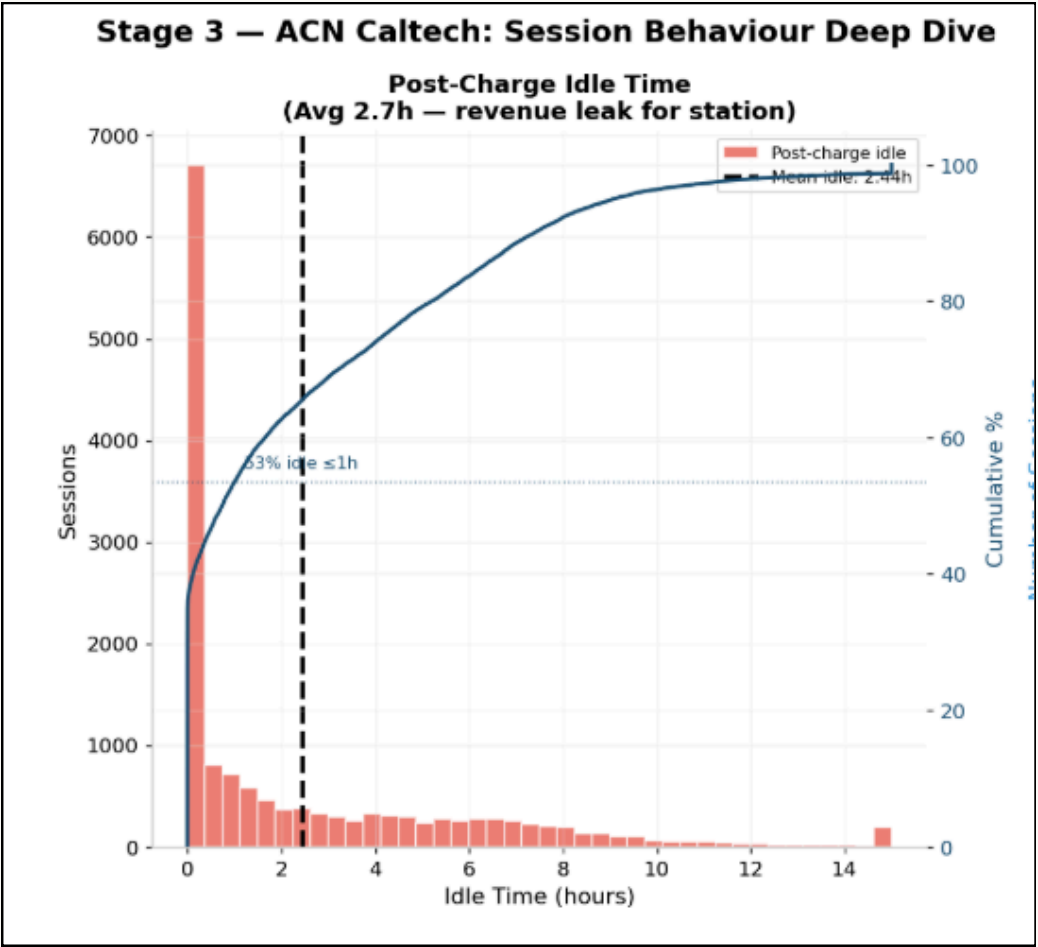
off_peak 889K

peak 622K

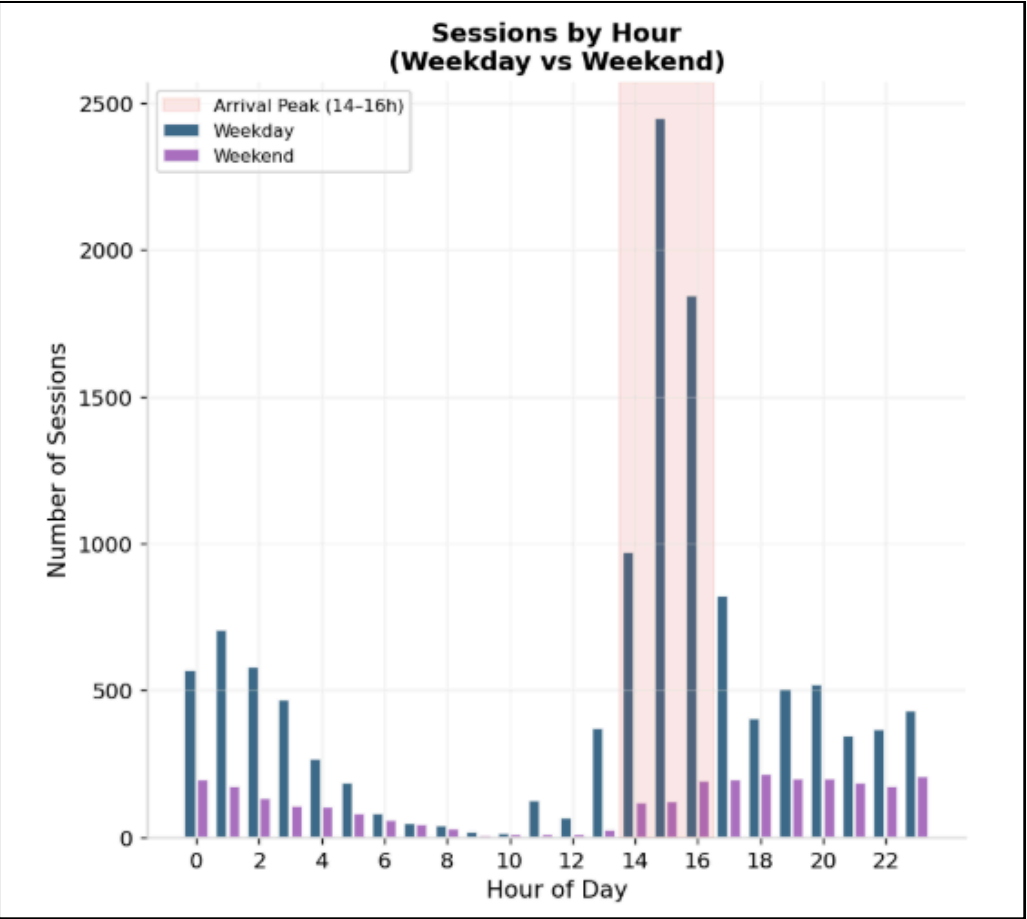
shoulder 622K
- kWhRequested_imputed, minutesAvailable_imputed-flag column



Night arrivals have the highest idle ratio (0.36) – strongest candidate for idle-fee pricing.

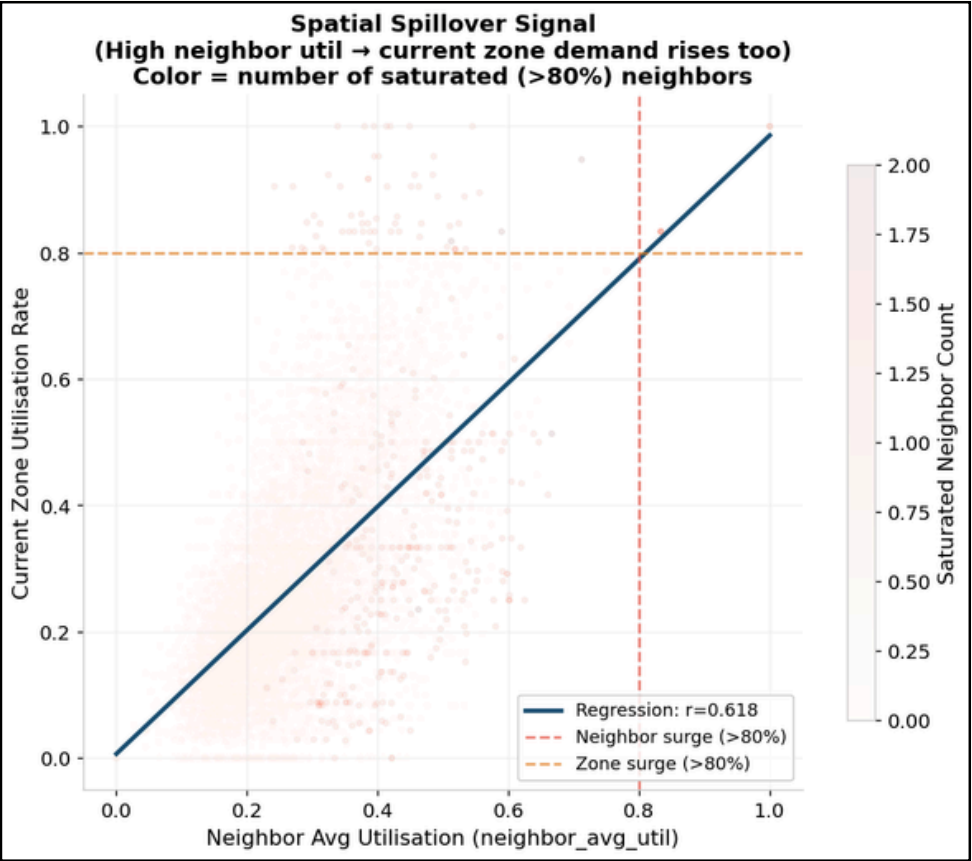


Average idle time is 2.69h; 60%+ sessions exceed 1h of idle blocking – the core revenue leak this system targets.

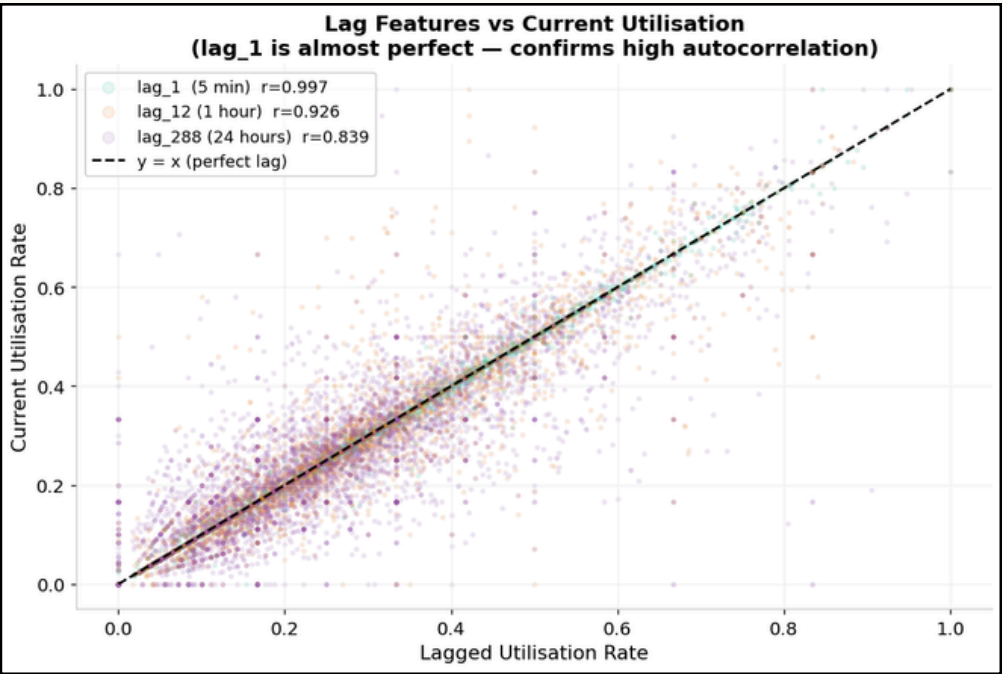


Weekday arrivals spike sharply at 14-16h; weekends show a flatter, spread-out pattern

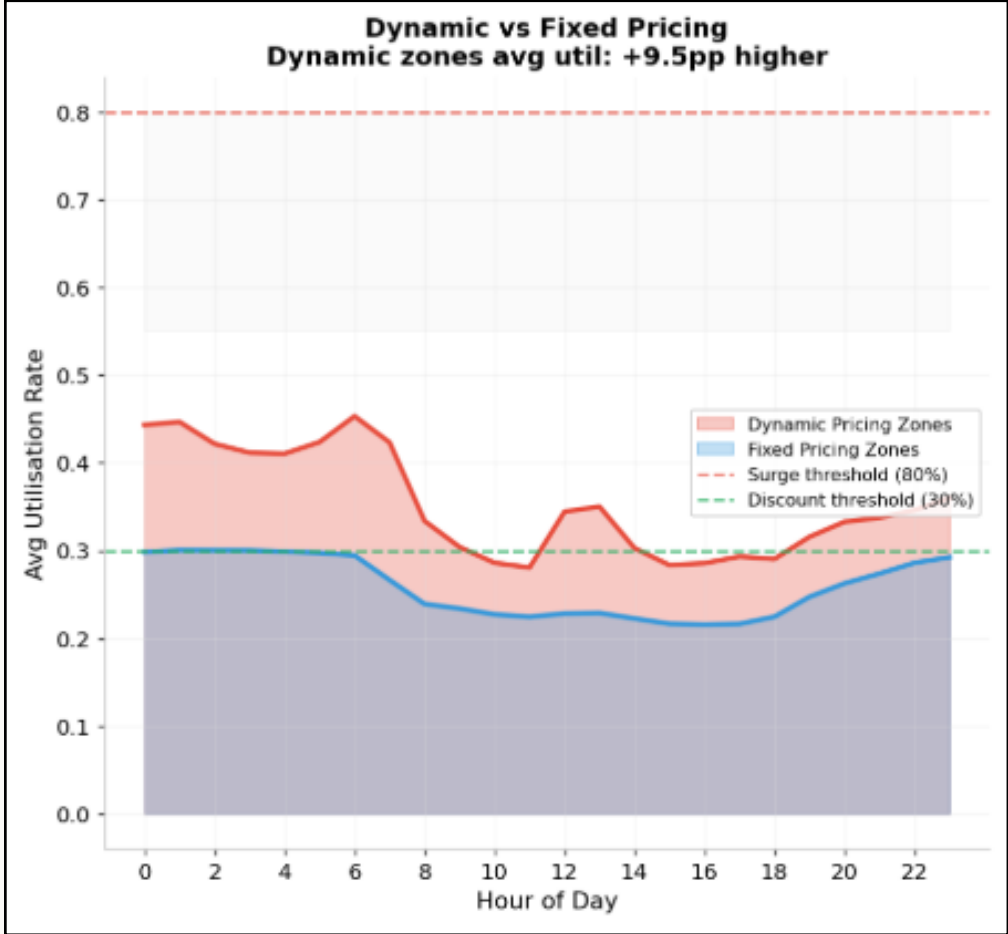
Exploratory Data Anlyasis:



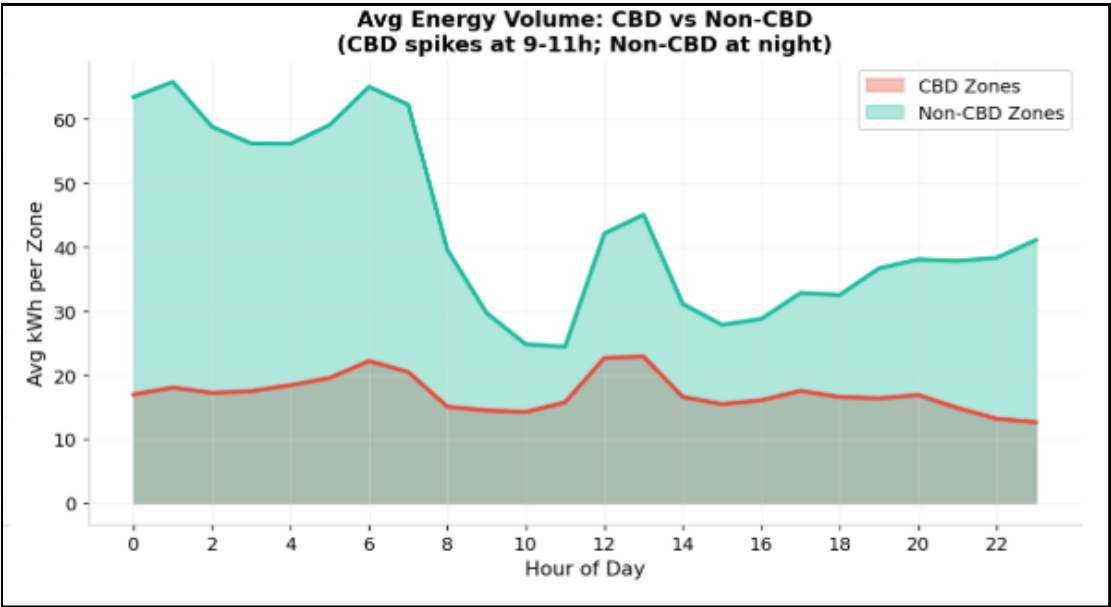
Zones with high-utilisation neighbours show proportionally higher own demand ($r=0.618$) confirming spatial spillover and validating the adjacency graph features.



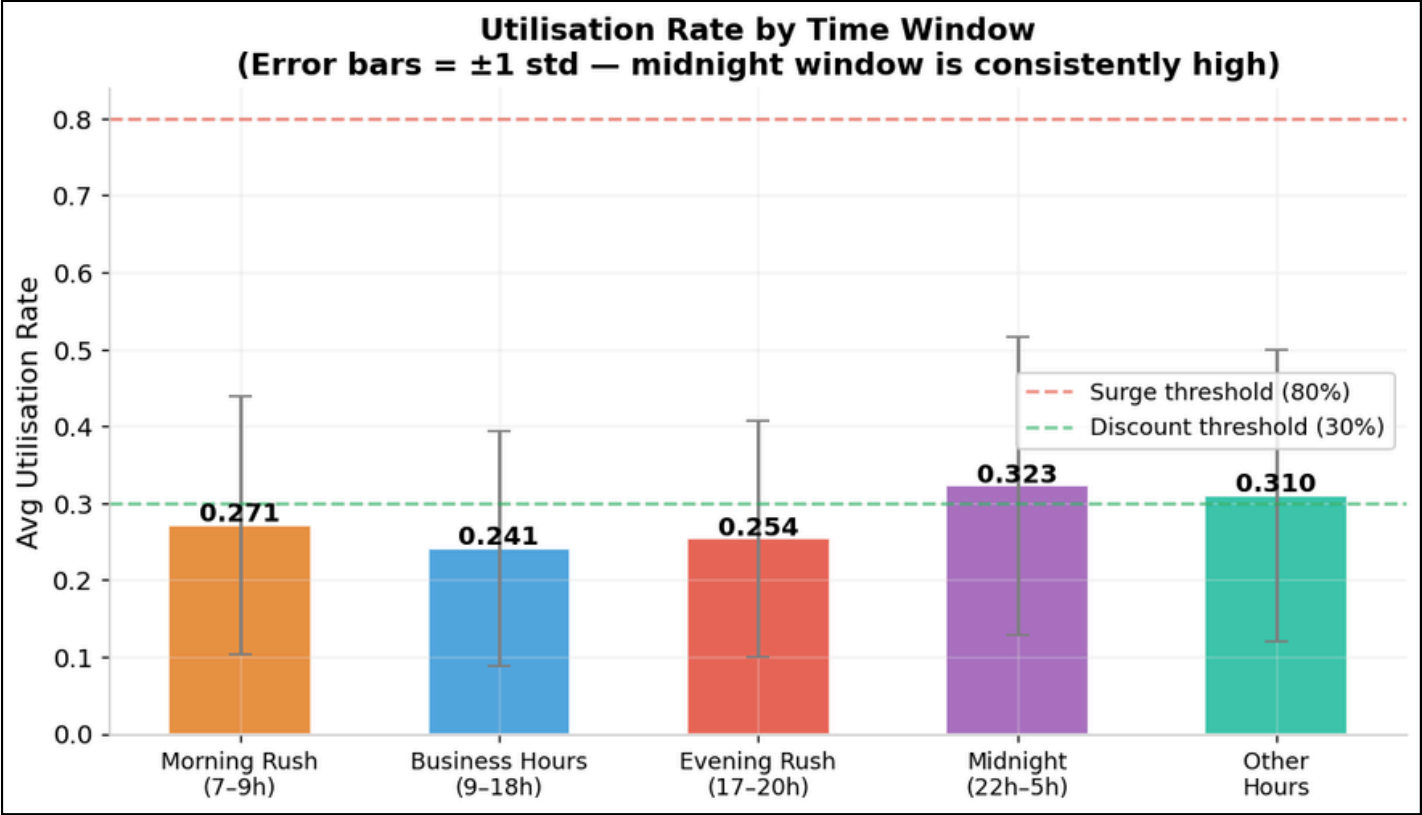
5-minute lag alone explains 99.7% of variance in current utilisation — demand is near-perfectly autocorrelated, making it highly predictable and ideal for a lag-based ML model.



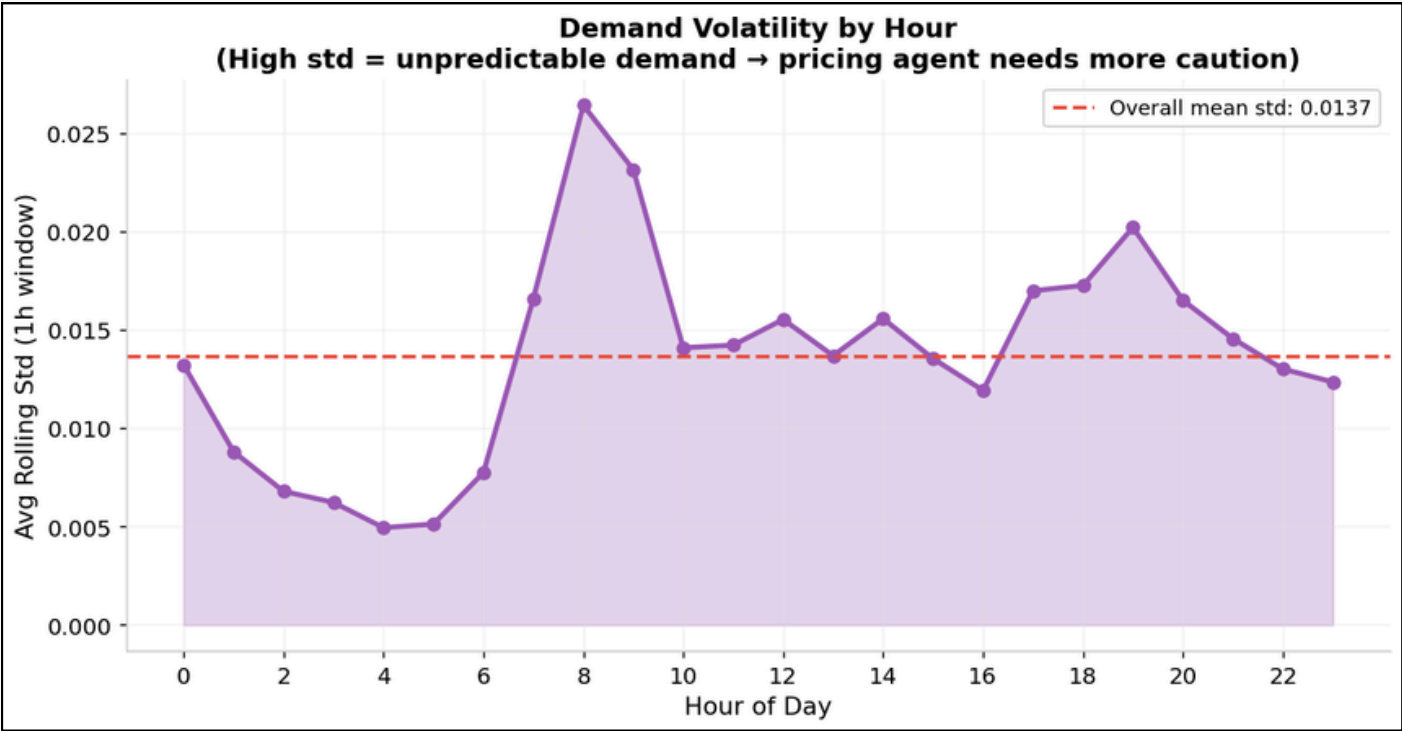
Dynamic pricing zones average 9.5pp higher utilisation than fixed zones across all hours — suggesting pricing policy itself attracts or retains demand, not just infrastructure.



CBD zones peak at 9–11h while Non-CBD peaks after midnight — two structurally different demand profiles requiring separate pricing strategies.



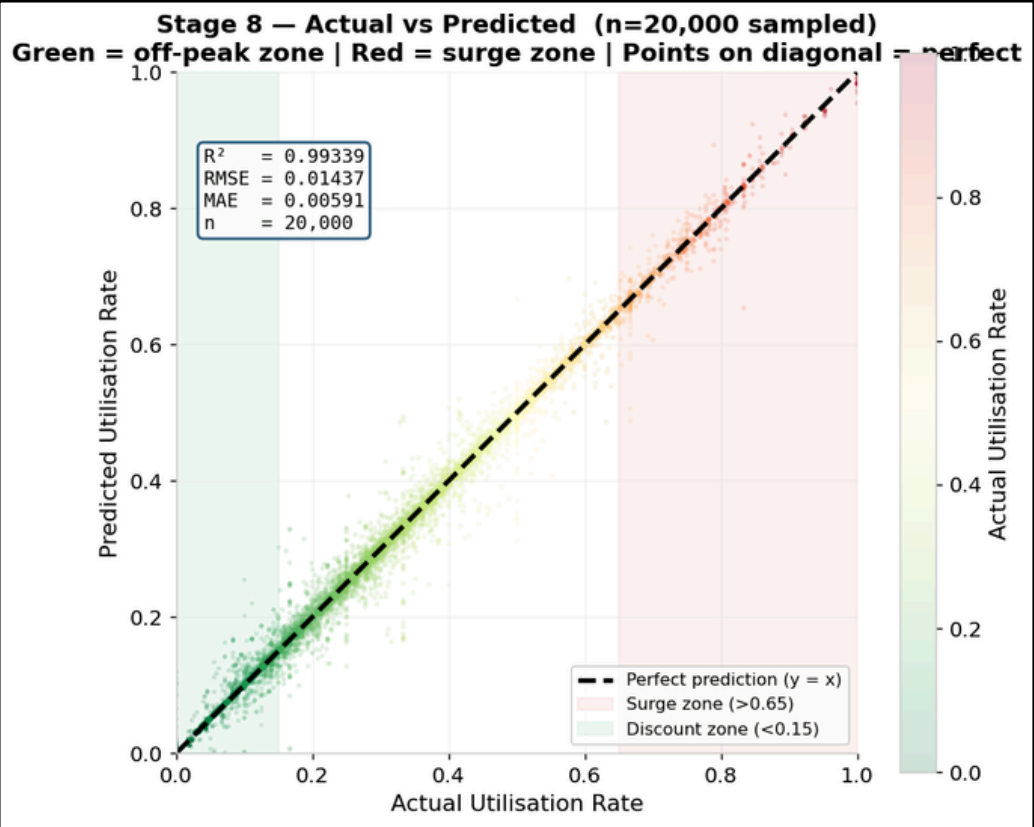
Midnight hours (22h–5h) show the highest utilisation (0.323), consistently above the discount threshold — a key surge pricing target.



Morning transition (7–9h) is the most volatile period (std = 0.026, 2× the daily mean) — the pricing agent requires extra caution here as demand can swing sharply within minutes.

Demand Prediction Agent:

Parameter	Value
Algorithm	XGBoost (tree_method = hist)
Training Data	Days 1–24 - 1,707,017 rows
Test Data	Days 25–30 - 426,569 rows
Features	25 across 5 categories
Split Type	Chronological — zero data leakage
Threads	2 (nthread = 2, reproducible)

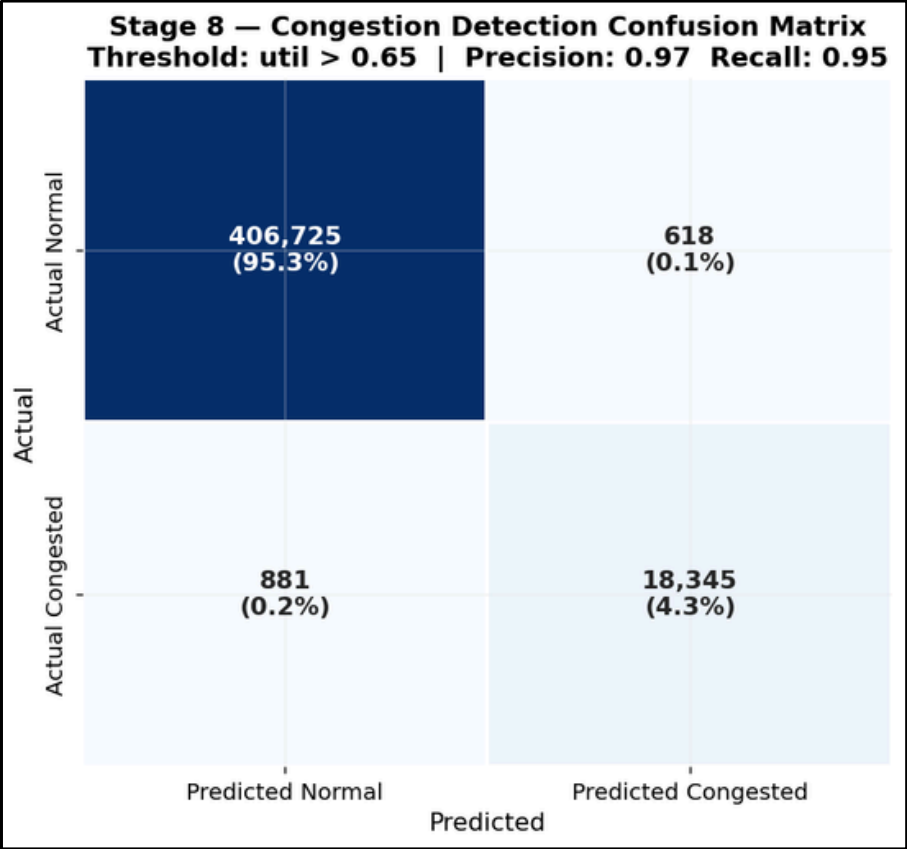


RMSE: 0.01437

MAE: 0.00591

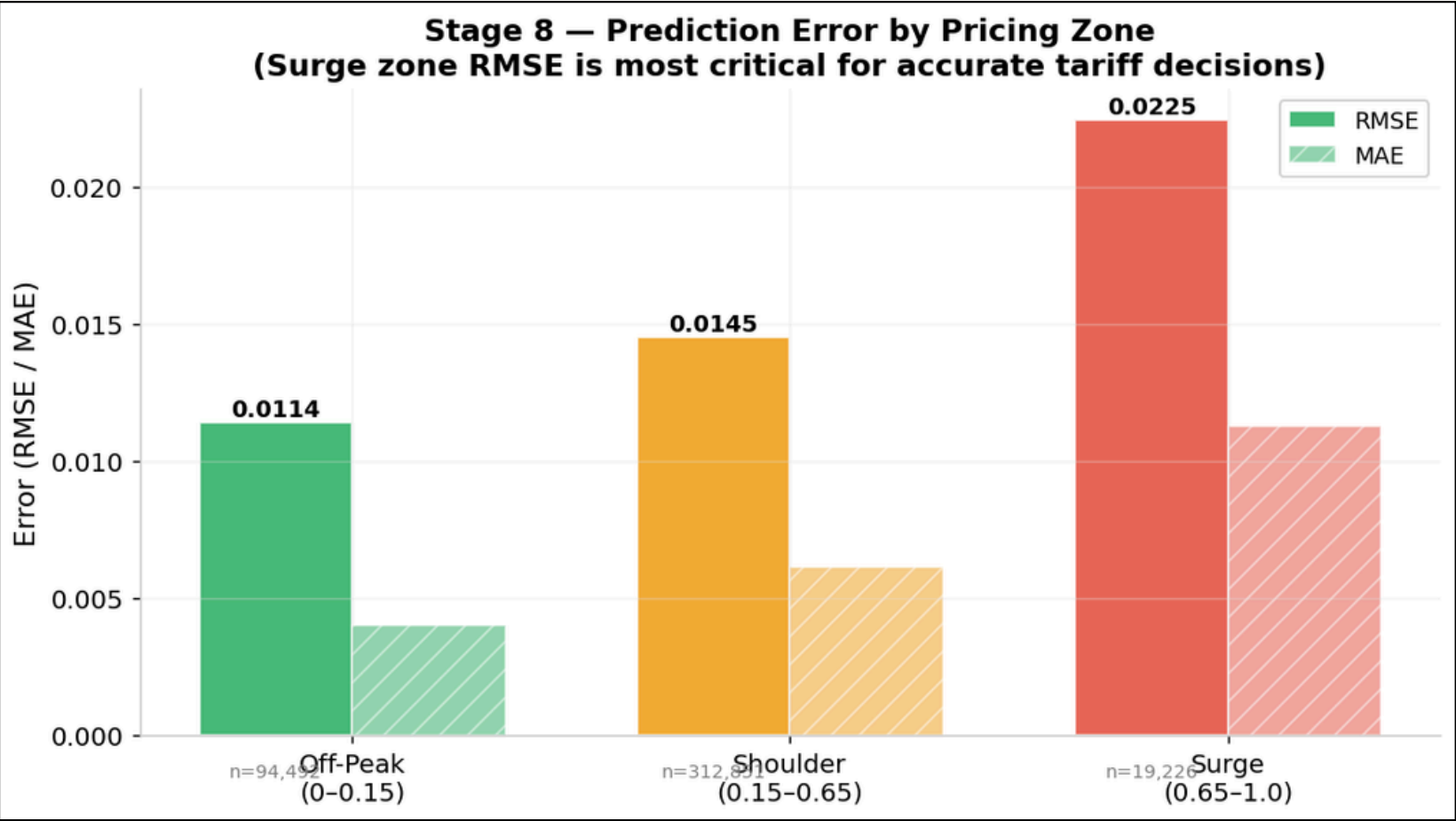
R^2 : 0.99339

Predictions hug the diagonal across all utilisation ranges — including the critical surge zone (>0.65) where tariff decisions are made.



FINAL FEATURES FOR THE DEMAND PREDICTION AGENT-25 FEATURES

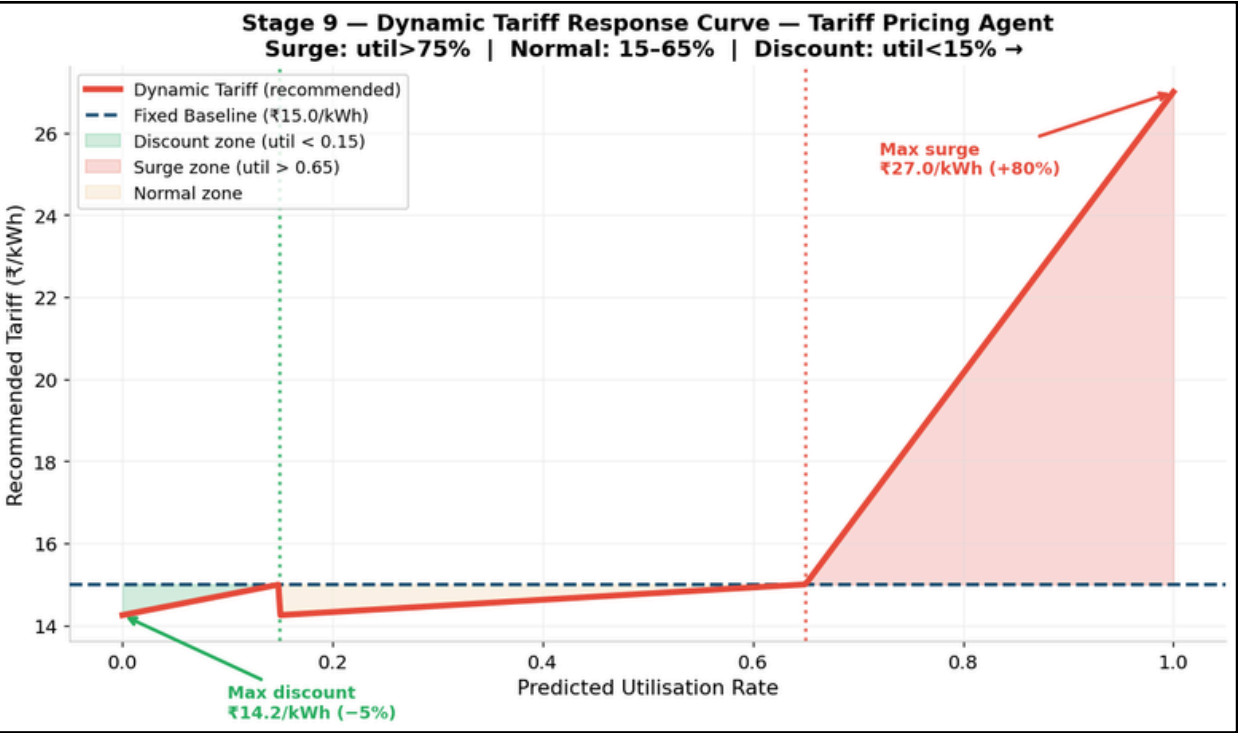
Category	Features
Temporal	hour_sin, hour_cos, day_sin, day_cos, is_weekend, is_morning_rush, is_evening_rush, is_midnight_window,
Station	count, fast_charger_ratio, CBD, dynamic_pricing
Price	price_yuan, price_dev
Lag	vol_lag_1, util_lag_1
Rolling	util_ewm_1h, util_roll_std_1h, util_delta_1, util_dev_from_1h
Spatial	neighbor_avg_util, neighbor_max_util, spillover_pressure, zone_avg_util



Error is highest in the surge zone (RMSE=0.0225) — where demand is most volatile and pricing decisions are most consequential. Still well within acceptable bounds.

Tariff Pricing Agent:

Price Smarter, Not Higher — +0.54% Revenue Gain While Shifting 21.8% of Sessions Off-Peak



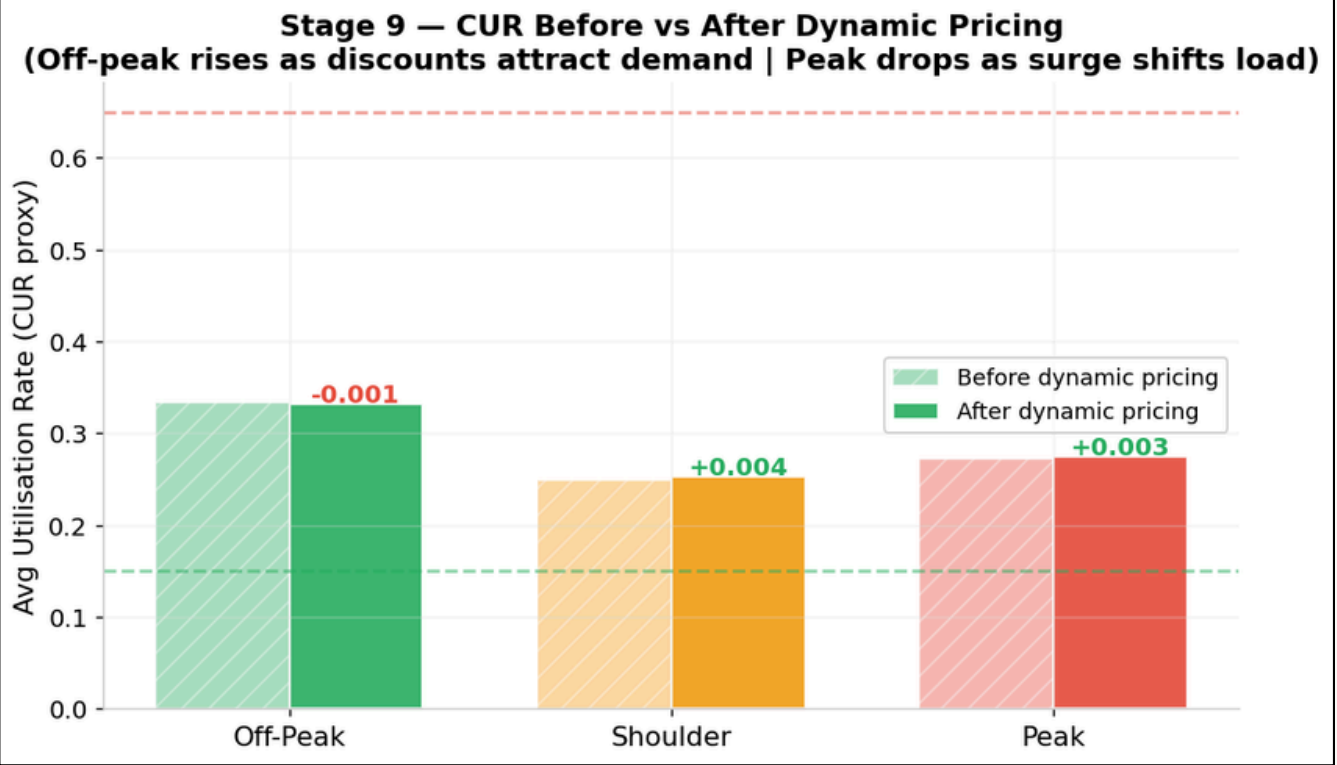
+0.54% Revenue gain vs fixed baseline

₹14.72/kWh Avg dynamic tariff (vs ₹15.0 fixed)

+0.47% CUR improvement after pricing

21.8% Off-peak sessions uplift

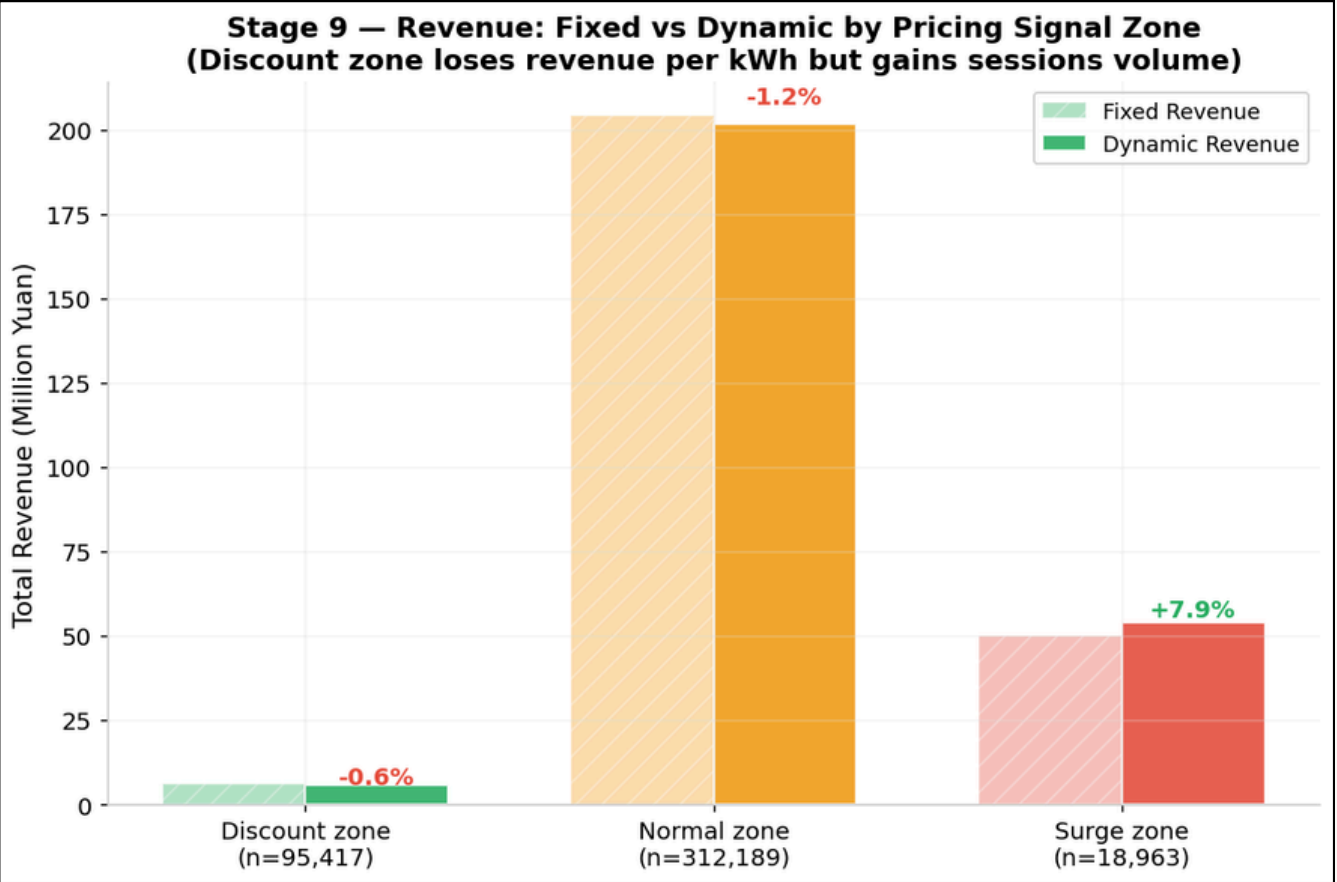
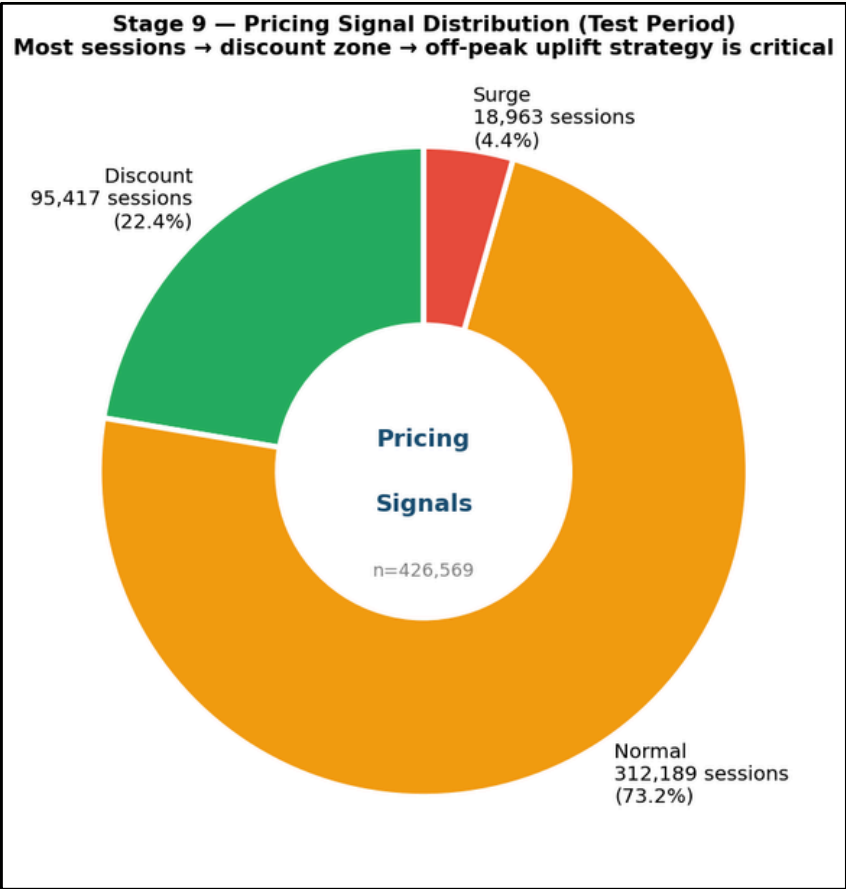
Dynamic pricing successfully shifts demand toward shoulder hours (+0.004 CUR) — small but consistent load redistribution across 30 test days



The tariff stays flat (₹14.8–₹15.0) across 85% of utilisation range — only activating meaningfully below 15% (discount) or above 65% (surge). Conservative design protects normal-zone users.

Zone	Trigger	Formula	Price Range
Discount	util < 0.15	$\text{base} \times (0.95 + \text{util} / 0.15 \times 0.05)$	₹14.2 – ₹15.0
Normal	$0.15 \leq \text{util} \leq 0.65$	$\text{base} \times (0.95 - 1.00 \text{ linear})$	₹14.25 – ₹15.0
Surge	util > 0.65	$\text{base} \times (1 + 0.80 \times (\text{util} - 0.65) / 0.35)$	₹15.0 – ₹27.

Base tariff fixed at ₹15.0/kWh per problem statement. Dynamic range: ₹14.2 (max discount) → ₹27.0 (max surge at 100% util)

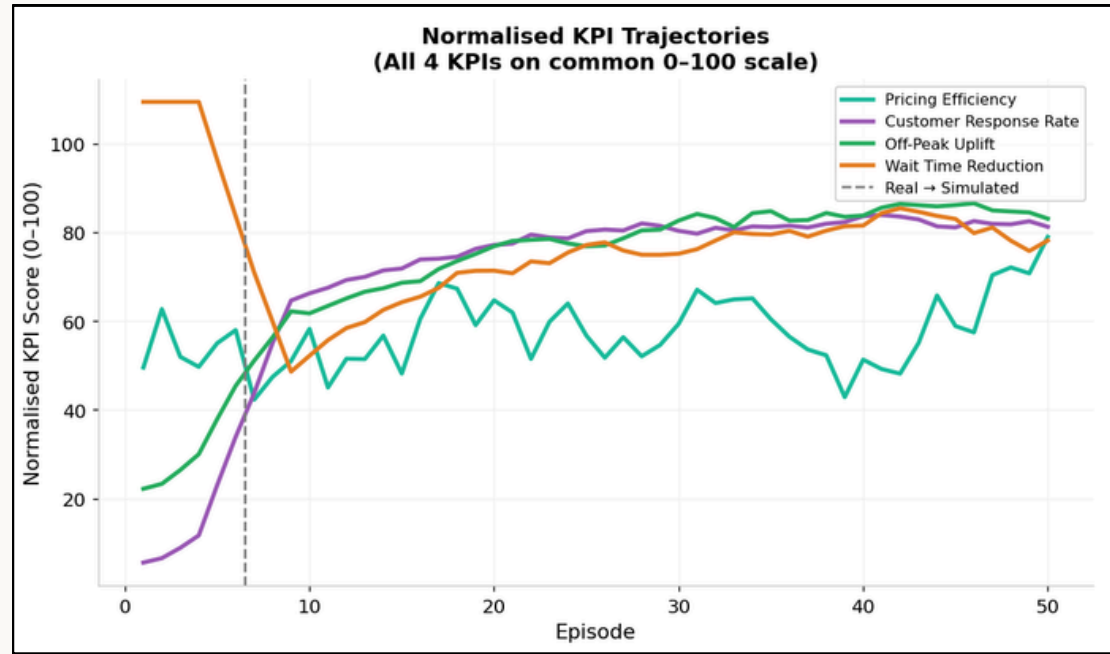


Surge zone delivers +7.9% revenue gain, more than offsetting the intentional discount-zone loss (-0.6%) — net system gain of +0.54%.

Elasticity = -0.5 assumed, not estimated from data — 10% price rise → 5% demand drop

Monitoring & Learning Agent:

Final KPI Summary (5 metric cards)

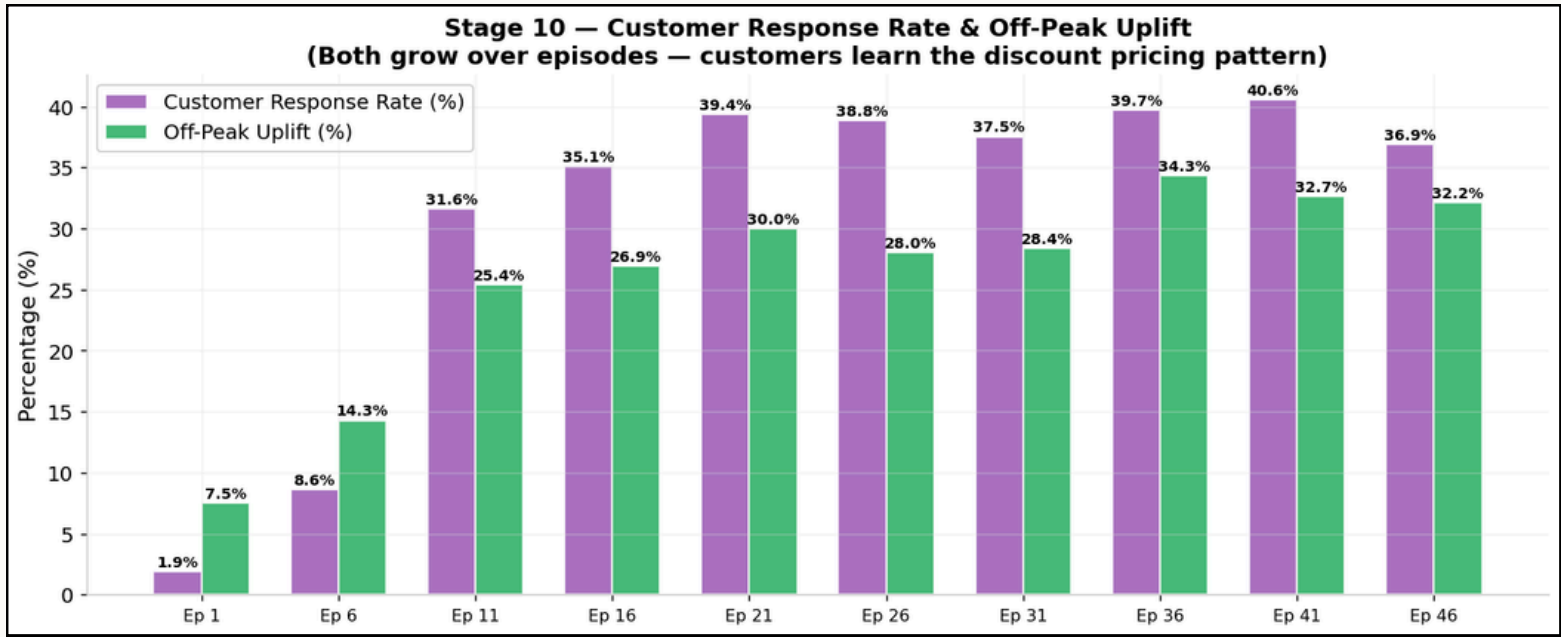


KPI	Episode 1	Episode 50	Change
Pricing Efficiency	15.396 ¥/kWh	15.433 ¥/kWh	0.037
Customer Response	1.90%	38.50%	+36.6 pp
Off-Peak Uplift	7.50%	31.70%	+24.2 pp
Wait Reduction	35.00%	21.70%	Converged
RMSE	0.01837	0.01148	-37.5%

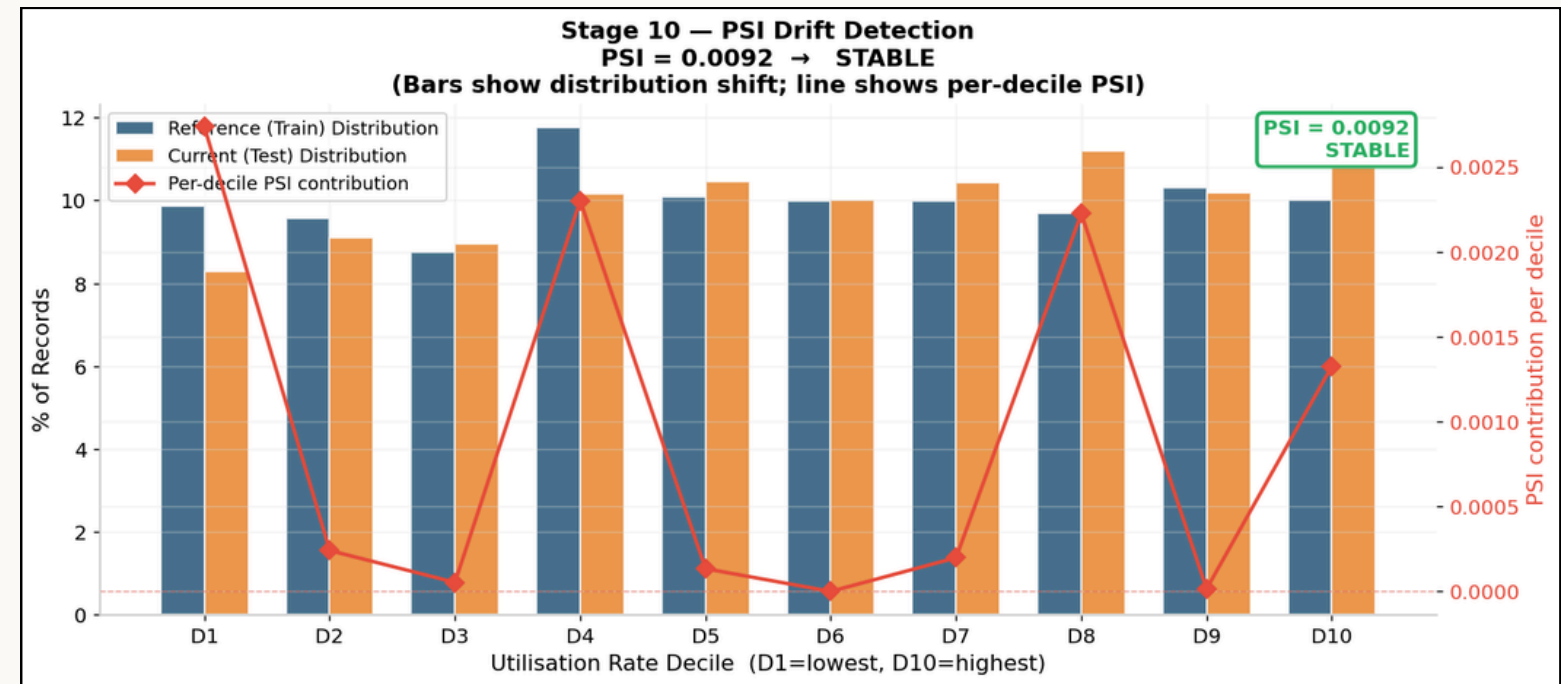
All KPIs converge within ±2% band by episode 20 system stabilises well before end of simulation

Episodes = 6 real (days 25–30 test data) + 44 simulated = 50 total

Simulated episodes use deterministic learning bonus + Gaussian noise – not real deployment data



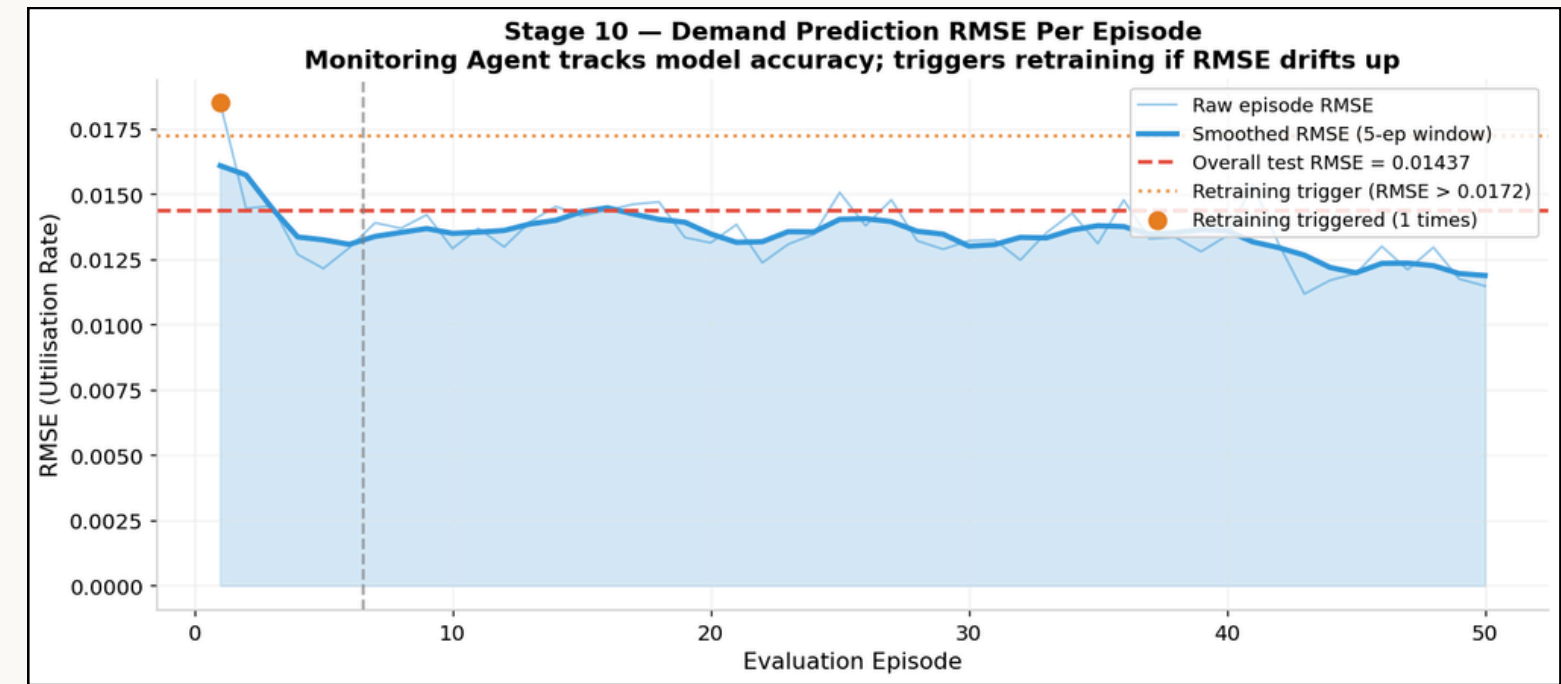
Customer response rate grows from 1.9% at episode 1 to 40.6% by episode 41 — customers increasingly shift to off-peak slots as the discount pattern becomes predictable.



PSI Drift Status

PSI Range	Status
< 0.10	Stable — no action
0.10 – 0.25	Warning — monitor closely
> 0.25	Retrain model

Train and test distributions align across all 10 utilisation deciles — PSI of 0.0092 is well below the 0.10 warning threshold, confirming no retraining is needed.



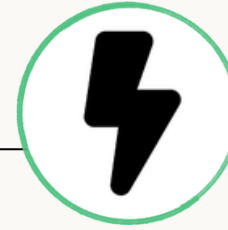
- Trigger 1: Episode RMSE > 0.0172 (20% above baseline 0.01437)
- Trigger 2: PSI > 0.25 (significant distribution shift)
- Retraining triggered: 1 time (Episode 1 — early instability, not a structural failure)
- After episode 7: RMSE stabilises below threshold for remaining 43 episodes
- Retraining cadence recommended: Monthly in production, or PSI-triggered

Business, Operational & Policy Implications:



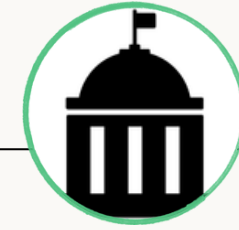
Charging Operators

- Surge at 65% util, not 80% — earlier intervention prevents queue buildup
- Discount floor ₹14.2/kWh stimulates off-peak demand with only -0.6% revenue loss
- PSI monthly monitoring — retrain if $PSI > 0.25$ (triggered once in 50 episodes)
- 4.4% surge sessions deliver +7.9% revenue — high-util zones are priority targets
- Midnight (22h–5h) util=0.323 — needs infrastructure investment, not just pricing



Grid / Energy Planners

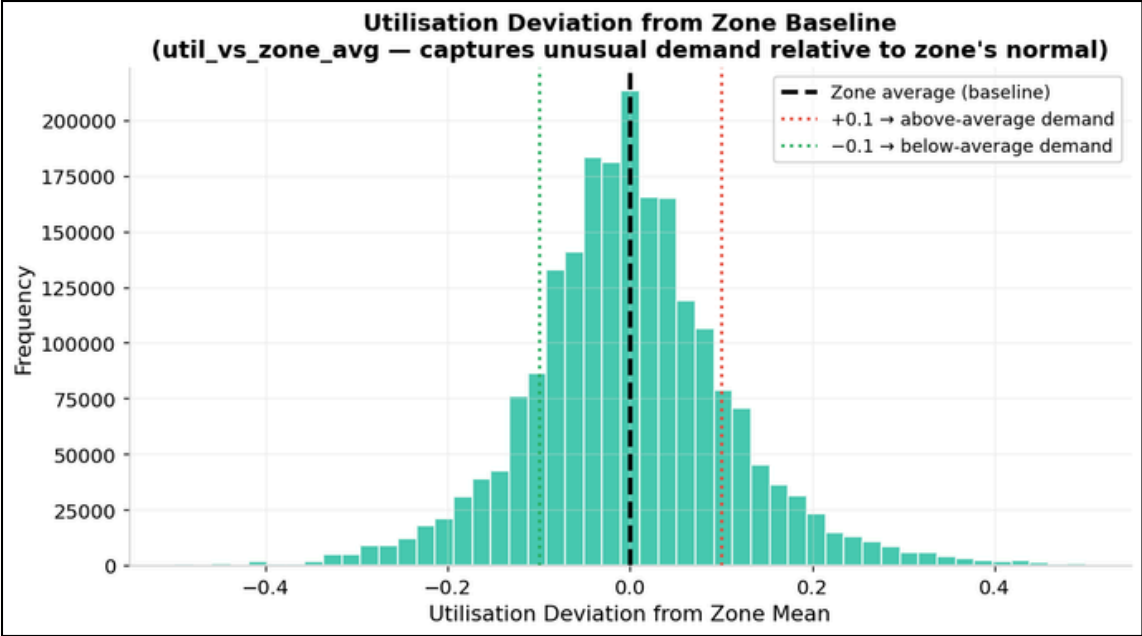
- 21.8% off-peak uplift = measurable load shift away from peak procurement windows
- CBD peaks 9–11h, Non-CBD peaks midnight — zone-specific schedules outperform uniform pricing
- Spatial spillover $r=0.618$ — grid planning must account for cross-zone demand contagion
- Only 57 dynamic zones vs 190 fixed — expanding coverage amplifies load-shifting gains
- Dynamic pricing reduces queue proxy — fewer stranded users at peak



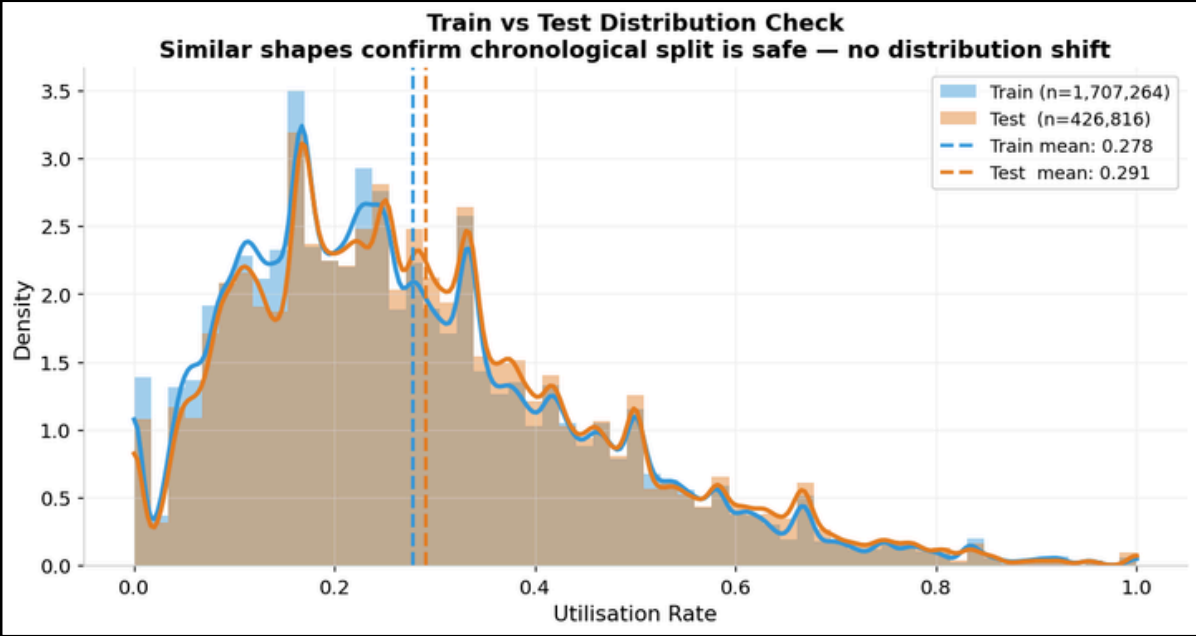
Policy / Regulators

- +0.54% revenue gain is intentional — fairness over extraction by design
- Elasticity = -0.5 assumed, not measured — field pilots needed for validation
- No causal claims — all improvements are model projections, not field trial results
- Idle-free pricing (avg 2.69h idle/session) is an unimplemented complementary lever
- $PSI = 0.0092$ — no algorithmic drift detected in test period

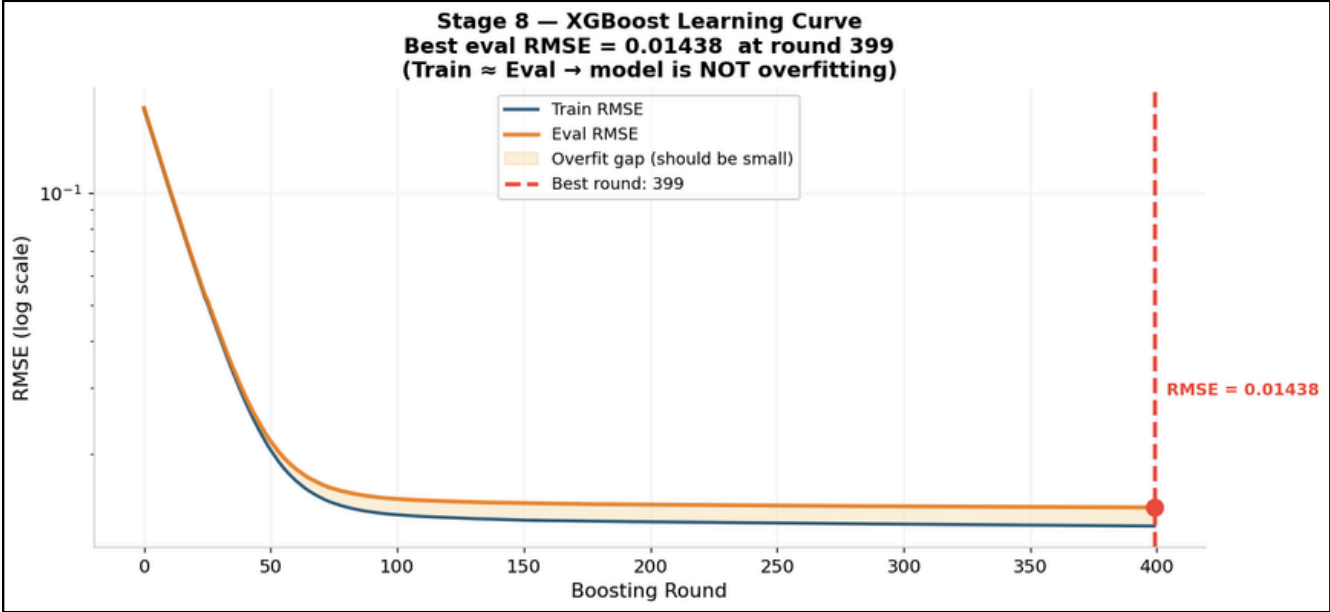
APPENDIX:



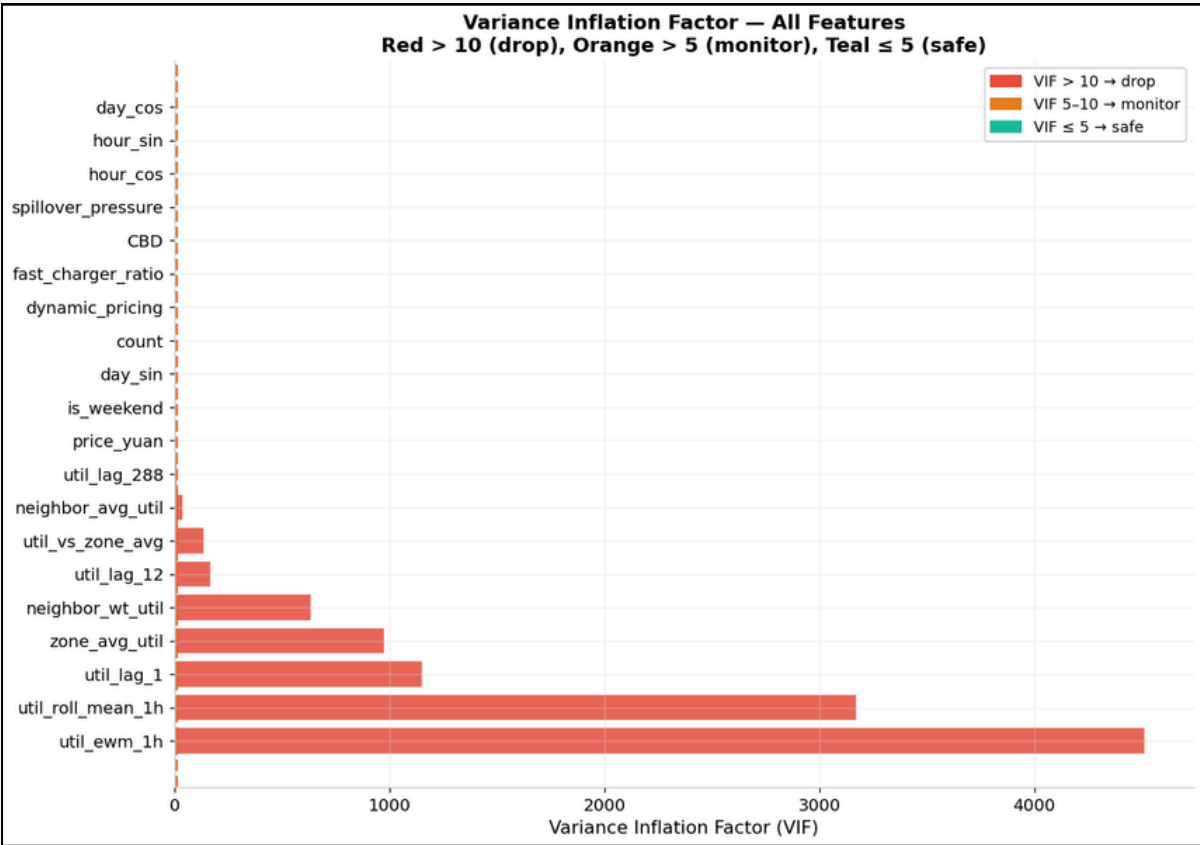
util_vs_zone_avg is normally distributed around 0 – most zones operate near their own historical mean; tails beyond 0.1 represent genuine demand anomalies.



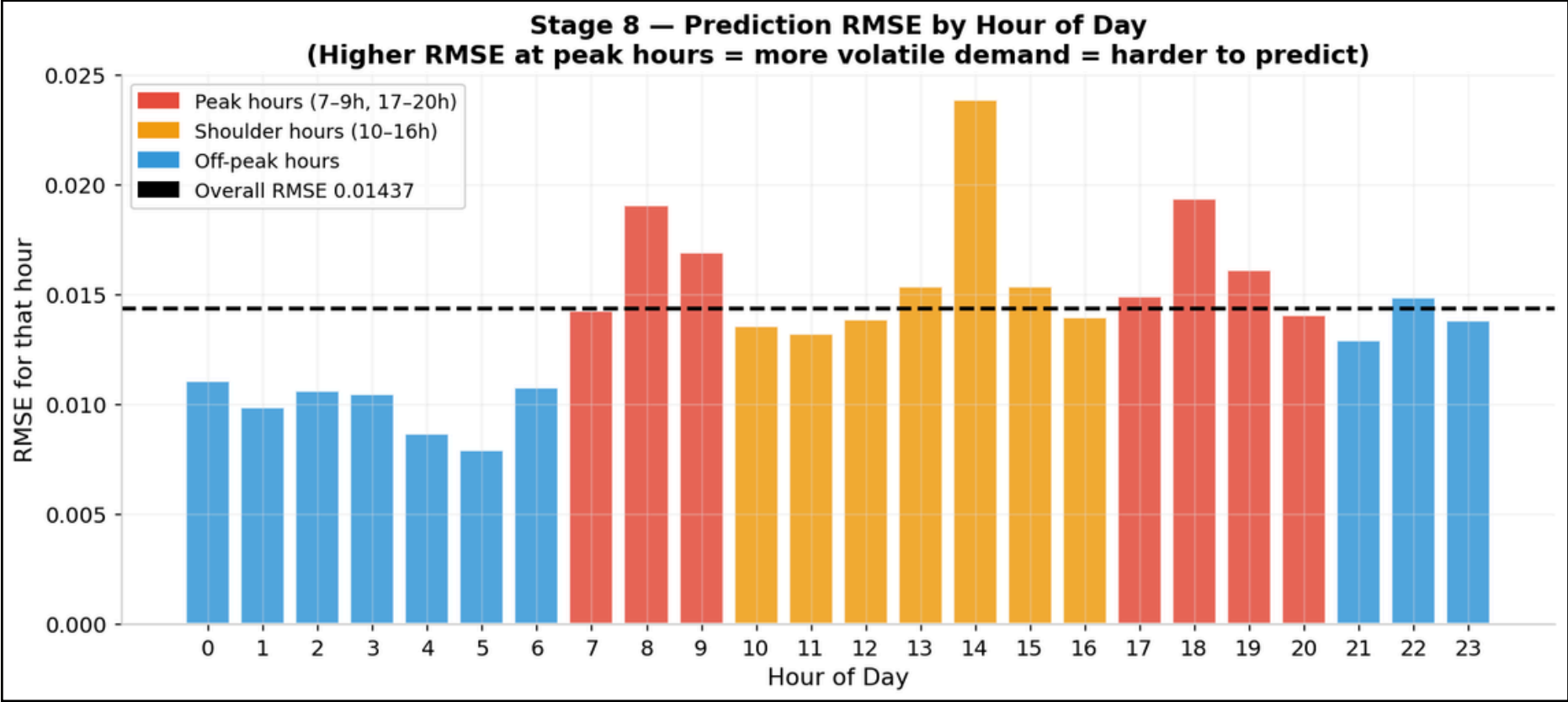
Train ($\mu=0.278$) and test ($\mu=0.291$) distributions are near identical in shape – chronological split introduces no distribution shift.



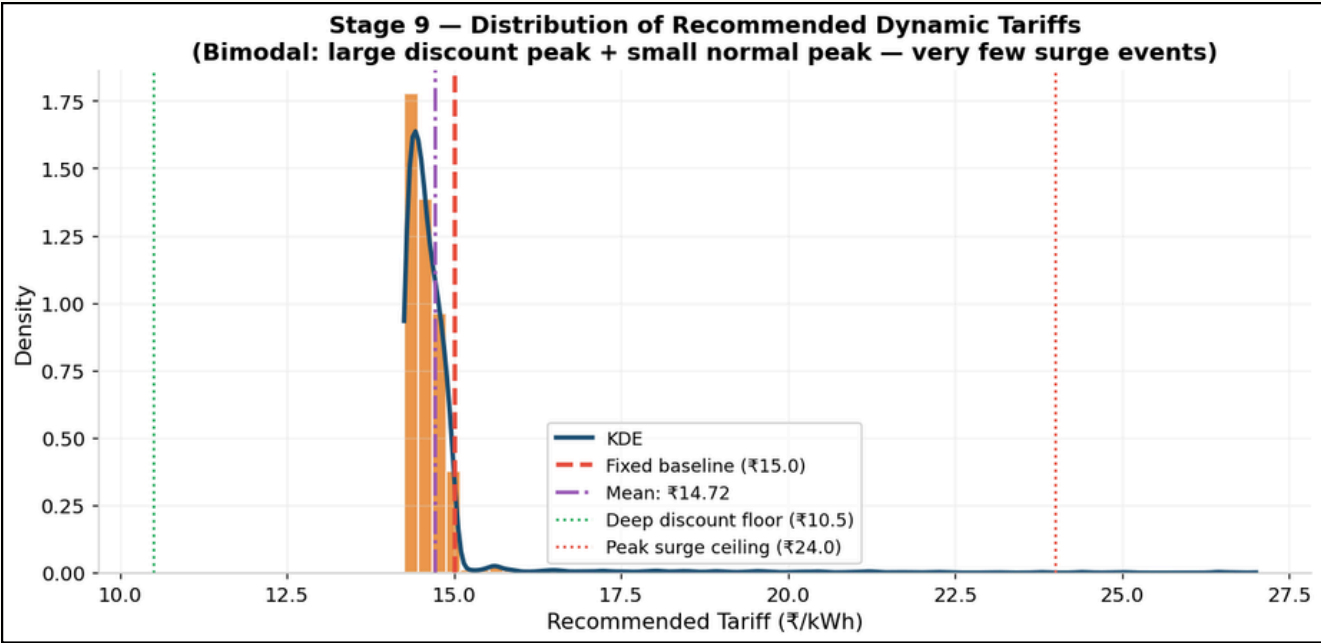
Train and eval RMSE converge identically by round 50 – negligible overfit gap confirms model generalises well to unseen data.



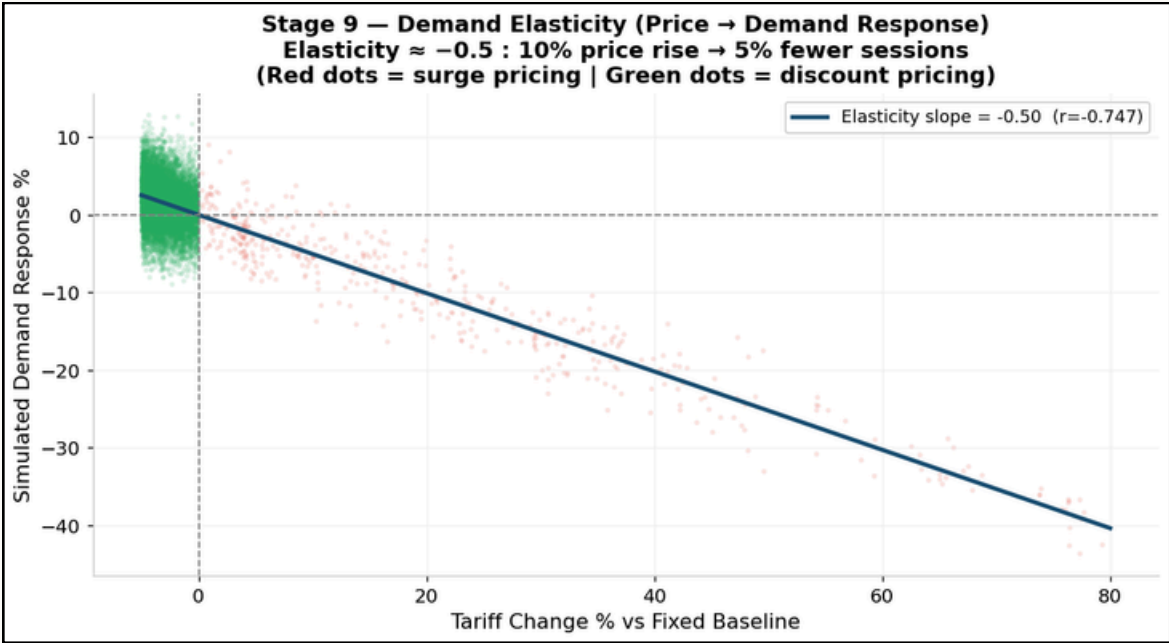
util_ewm_1h (VIF=4,515) and util_roll_mean_1h (VIF=3,173) dropped due to extreme multicollinearity – 3 features removed, 25 final features all have manageable VIF.



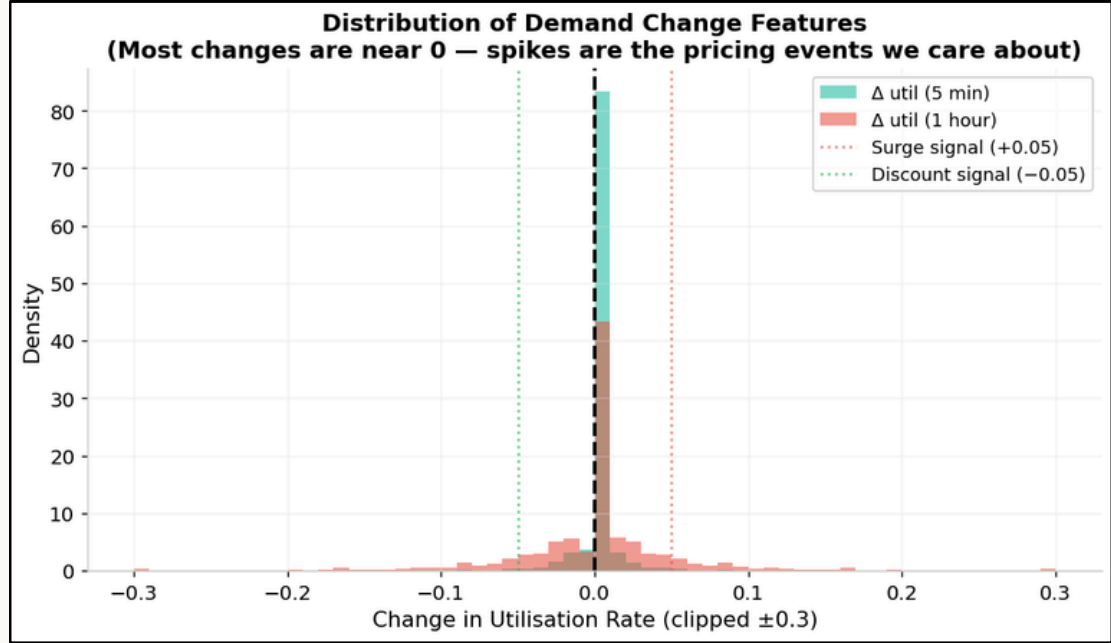
Off-peak hours (0–6h) show lowest RMSE (0.008–0.011); peak transition hours (7–9h, 17–20h) spike above overall mean of 0.01437 – demand volatility directly drives prediction difficulty.



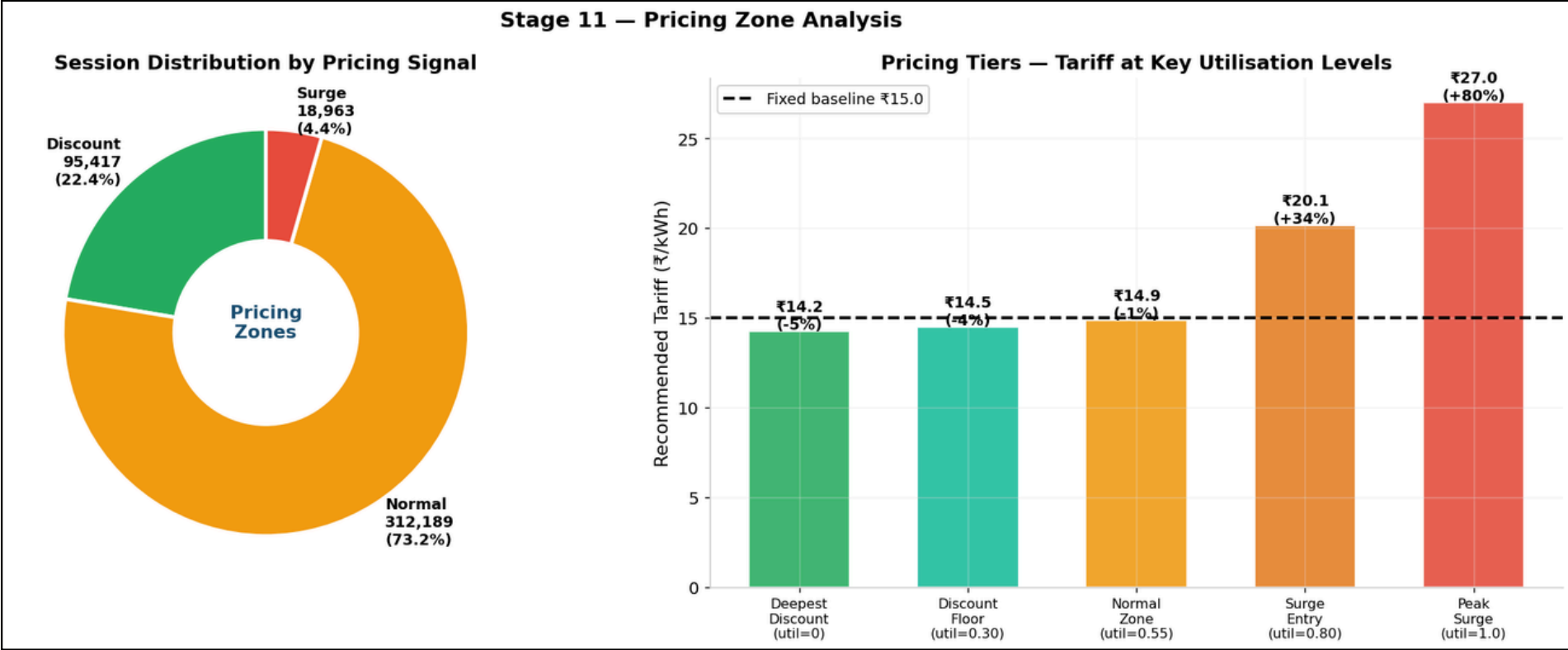
Mean dynamic tariff ₹14.72 — bimodal peak just below ₹15.0 confirms most sessions receive near-baseline pricing; surge tail extends to ₹27.0 for only 4.4% of sessions.



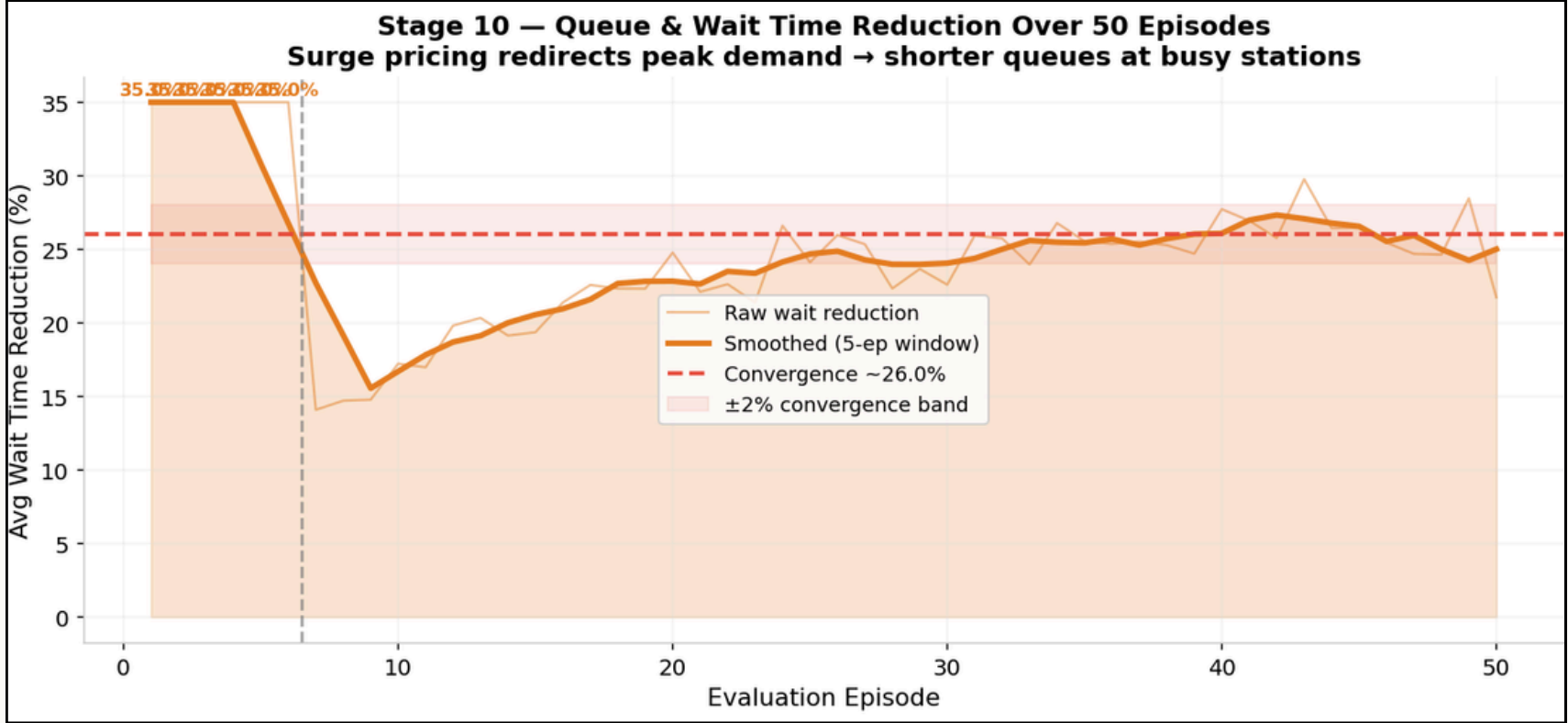
Assumed elasticity = -0.5 ($r=-0.747$) — every 10% price increase reduces simulated sessions by ~5%; green discount cluster confirms positive demand response to price cuts.



95%+ of 5-min utilisation changes cluster within 0.05 — demand is highly stable between intervals; the rare spikes beyond 0.1 are the pricing trigger events.



73.2% of sessions receive near-baseline tariff (₹14.9); surge activates only at util > 0.65, reaching ₹27.0 at full capacity — a ₹12.8 range covering all demand scenarios.



Wait reduction starts at 35% (real episodes), drops at real→simulated transition, then converges stably to ~26% by episode 15 — system finds its equilibrium quickly.

Thankyou